

Mahashri D

Software Engineer | AI / GenAI | Python | Data & Analytics

Mobile: +91-8524035315 Email: mahashridharmarajan182@gmail.com LinkedIn: linkedin.com/in/mahashrid

Location: Chennai, Tamil Nadu, India

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience at CGI building AI-powered code analytics and data-driven software solutions. Experienced in Python, SQL, React, Power BI, AST-based code analysis, LLM integration, and enterprise code-quality platforms using SonarQube, CAST, and Checkmarx. Developed intelligent code-analysis workflows, interactive dashboards, AI-powered code explanations, and automated code recommendations, with a strong foundation in Data Science and Machine Learning.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI / GenAI: LLMs, Prompt Engineering, AI-Powered Code Analysis, Code Explanation, Automated Code Recommendations

Data & Analytics: Pandas, NumPy, Data Analysis, Data Processing, Statistical Analysis

Machine Learning: Scikit-learn, Regression, Classification, Clustering

Web Development: React.js, HTML, CSS, REST APIs

Data Visualization: Power BI, Tableau, Matplotlib, Seaborn

Code Quality & Tools: SonarQube, CAST, Checkmarx, AST, IntelliJ Plugin Development

Databases & Cloud: PostgreSQL, AWS EC2, AWS S3

DevOps & Tools: Git, Docker, Jenkins, Jupyter Notebook

PROFESSIONAL EXPERIENCE

CGI

Aug 2023 – Present

Associate Software Engineer

RightCoder – Code Quality Analytics Platform

- Contributed to an enterprise code-quality analytics platform integrating **3 code-quality tools** – SonarQube, CAST, and Checkmarx – to consolidate code quality, security, and project-level risk analysis.
- Developed Python-based data-processing workflows using Pandas and NumPy to analyze code-quality metrics, issue severity distributions, and project-level quality indicators.
- Built interactive React and Power BI dashboards to consolidate code-quality metrics, risk insights, severity trends, and project analytics, reducing dependency on manual analysis workflows.
- Contributed to **2 developer-facing interfaces** – an IntelliJ plugin and web interface – for accessing analysis results within developer workflows.
- Developed an **AI-powered chatbot** to provide faster access to code-quality information and analysis results.

Signet – AI-Powered Code Analysis & Recommendation System

- Developed a Python-based AST code analyzer to identify code complexity, nesting depth, and code patterns for automated code-quality analysis.
- Performed data processing and analysis using Python (Pandas, NumPy) to derive code-quality metrics, severity distributions, and identify quality trends.
- Built interactive React and Power BI dashboards to visualize severity distributions, quality metrics, trends, project insights, and risk hotspots.
- Implemented LLM-powered code explanations and automated fix recommendations to help developers understand and remediate code-quality issues.
- Enhanced the code-analysis portal with project insights, scan comparison, findings, quality profiles, rules, quality gates, scan history, and Git/SCM analysis workflows.
- Improved React frontend performance for data-intensive visualizations using caching, in-flight request deduplication, controlled concurrency, virtualization, pagination, and optimized component rendering.
- Containerized application components using Docker and integrated build and deployment workflows with Jenkins.

EDUCATION

MSc Data Science

2025

Manipal University, Karnataka

Bachelor of Computer Applications (BCA)

2023

Bishop Heber College, Tiruchirappalli, Tamil Nadu

CERTIFICATIONS

- Google Data Analytics Certificate** – Google
- Artificial Intelligence Certificate** – Microsoft Azure
- Machine Learning for All** – University of London
- Leadership Skills Certificate** – IIM Ahmedabad

ACHIEVEMENTS

- Received **Bronze Award – CGI (Apr 2026)** for contributions to AI-powered code quality platform development.
- Presented a research paper on “**Cloud Computing and its Role in IT Companies**” at the ICICS 2022 International Conference, Bishop Heber College, Trichy (March 2022).